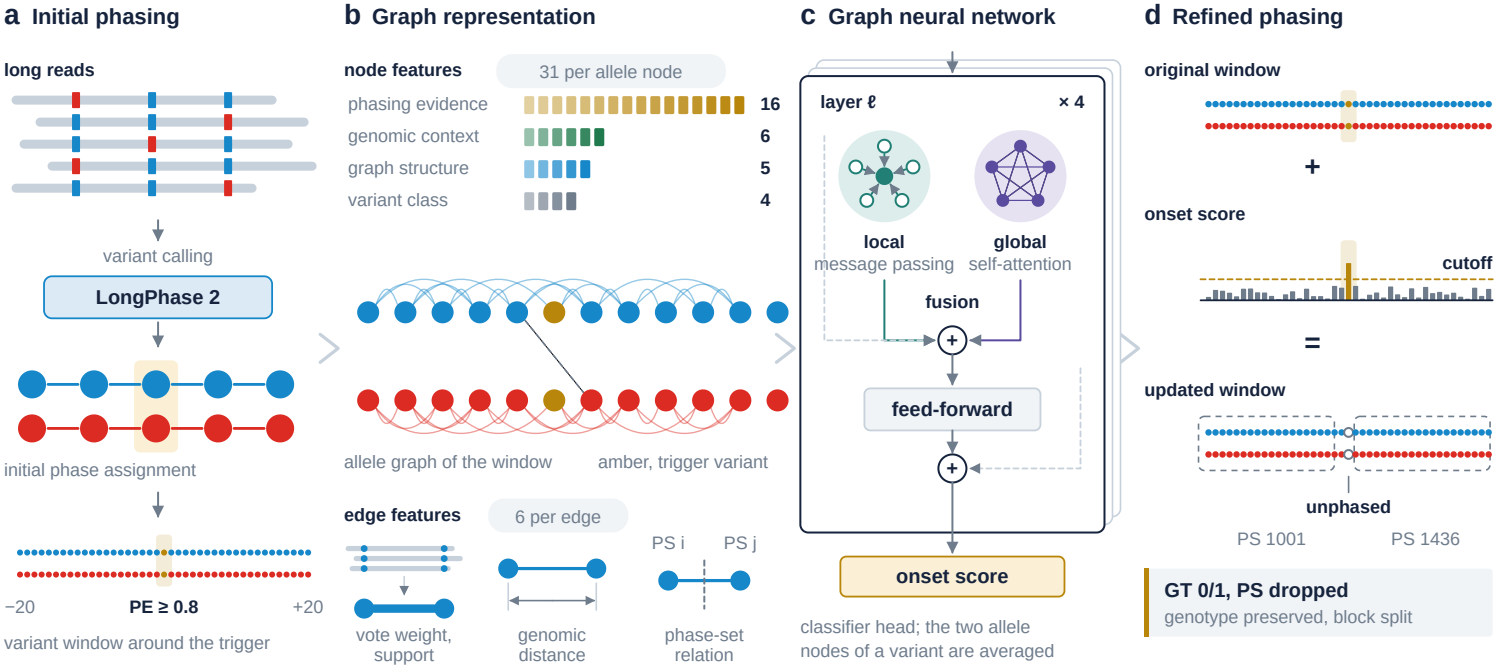

a Initial phasing
long reads
variant calling
LongPhase 2
initial phase assignment
−20
PE ≥ 0.8
+20
variant window around the trigger
b Graph representation
node features
31 per allele node
phasing evidence
16
genomic context
6
graph structure
5
variant class
4
allele graph of the window
amber, trigger variant
edge features
6 per edge
PS i
PS j
vote weight, support
genomic distance
phase-set relation
c Graph neural network
layer ℓ
× 4
local
message passing
global
self-attention
fusion
feed-forward
onset score
classifier head; the two allele nodes of a variant are averaged
d Refined phasing
original window
+
onset score
cutoff
=
updated window
unphased
PS 1001
PS 1436
GT 0/1, PS dropped
genotype preserved, block split